TakeSure · Confidence before taking something

An Ayurveda × Allopathy herb–drug interaction checker for Indian patients, doctors and the Ministry of Ayush

Domain: Health & Well-Being (AI4Impact secondary) · SDG 3 · SDG 9 · SDG 10

Team Bug Hunters

Dhriti Garg · Shiwani Goyal · Khushi Gupta

HACK@BVP 8.0 · Abstract Round

ONE PATIENT'S STORY — THE GAP TODAY, AND WHAT TAKESURE WOULD DO

1. **Mother** takes Thyronorm for thyroid and starts ashwagandha for stress.
2. **Nobody asks, nobody tells.** Her doctor never hears of it; the label says nothing.
3. **She photographs both.** TakeSure reads the names, confirms them and checks the pair.
4. **AMBER.** A cited source, timing advice, three questions. One tap sends the PDF to her doctor.

37% of hospital patients use allopathic and herbal medicines together

65% of herbal users do not know the dose, side effects or interactions

19% of Indian patients ever disclose herbal use to their doctor

1,473 ASU&H adverse-event reports received nationally since 2018

"The verdict comes from published data; AI only explains it."

1 THE PROBLEM

- **Two medical systems, zero cross-checking.** A thyroid patient on Thyronorm adds ashwagandha; a transplant patient on tacrolimus drinks giloy kadha; a cardiac patient on warfarin takes guggul.
- **Each pair can cause harm:** ashwagandha can raise thyroid hormone, giloy can oppose immunosuppression, guggul can increase bleeding.
- **Nobody asks, nobody tells.** The doctor rarely asks about herbs, the patient rarely mentions them, the chemist has no tool, the label says nothing.
- **Existing tools do not cover it.** Medscape, Drugs.com and 1mg check only allopathic × allopathic pairs; Western herb databases are English-only, paid and do not know Indian brands like Himalaya Ashvagandha or Dabur Giloy Ghanvati.

2 PROPOSED SOLUTION

- **Scan both.** The patient photographs the allopathic prescription and the Ayurvedic product label (or types the names); every name is confirmed before checking.
- **Match Indian names.** Brand → generic (Thyronorm → levothyroxine) and product → herbs (Chyawanprash → amla, ashwagandha…) with Sanskrit / Hindi / Latin synonyms.
- **Decide from data.** A rule engine checks every herb–drug pair against a curated, cited knowledge base: GREEN / AMBER / RED / NOT IN DATABASE, with timing advice.
- **Explain simply.** AI rewrites only the matched evidence into plain Hindi or English, with sources and three questions to ask the doctor.
- **Report in one tap.** A one-page PDF for the doctor and a pre-filled suspected-ADR report in the Ministry of Ayush format for the Ayush Suraksha portal.

3 APPROACH — PROJECT WORKFLOW

Prescription photo allopathic medicines

Product photo Ayurvedic label / brand

› **1 · OCR** read both images; user confirms names › **2 · Match** brand → generic; product → herbs; synonyms › **3 · Knowledge base** cited herb–drug table; grade A/B/C › **Rule engine** decides severity; AI does not › GREEN / AMBER / RED / NOT IN DB › **4 · AI explains** simple Hindi/English + 3 doctor questions › **Doctor PDF** verdict + sources / **ADR report** Ministry of Ayush format

Verdict rules: Worst row wins; an unknown pair is never shown as green; grade-C evidence caps at AMBER; the answer is identical with the AI switched off.

KNOWLEDGE BASE — SAMPLE ROWS

Herb	Drug	Verdict	What happens	Grade	Source
Ashwagandha	Levothyroxine	AMBER	May raise thyroid hormone	B	Sharma 2018, PMID 28829155
Guggul	Warfarin	RED	May increase bleeding risk	B	MSKCC About Herbs
Giloy	Tacrolimus	RED	May oppose immuno-suppression	C	Nagral 2021, LiverTox
Triphala	Oral medicines	GREEN	Keep a 2–3 h gap	C	Ayurvedic Pharmacopoeia

Sample rows of the curated knowledge base (the table the verdict comes from).

STACK & DATA

Next.js + TypeScript PWA (Hindi/English UI) · FastAPI + Pydantic REST API · PostgreSQL in production, SQLite offline · Tesseract OCR · RapidFuzz matching (≥85 accept, 70–85 ask the user) · Groq LLM, JSON-schema output, temperature 0 · ReportLab PDFs. No model training; runs fully offline on a laptop.

40–60 high-frequency herb × drug pairs (15 common herbs × 12 drug classes) curated from PubMed, NIH LiverTox, MSKCC About Herbs, WHO monographs and the Ayurvedic Pharmacopoeia of India, graded on the Stockley's model. Decision support only: "ask your doctor", never diagnosis or dosing.

4 EXPECTED IMPACT

Patients & caregivers

A sourced answer before the first dose, in their own language, instead of silence or a guess. Free for citizens.

Doctors, vaidyas & chemists

Finally learn what patients actually take; a QR on the chemist's bill or an API inside hospital and telemedicine systems makes the check routine.

Ministry of Ayush

Structured, reproducible ADR reports and real co-use data from the catchment of its 99 peripheral pharmacovigilance centres.

Novelty

Nothing public today checks an Ayurvedic product against an allopathic prescription. TakeSure builds the missing bridge table: Indian brands × Ayurvedic herb synonyms × graded evidence, open and auditable. Honest by design: grades, timing gaps instead of alarms, "not in database" instead of guesses.